Suman Sourav
Jonathan J. Huerta y Munive

AALBORG UNIVERSITY
DENMARK

Models and Tools for Cyber-Physical Systems
Exercise sheet 6
WITH SOLUTIONS

Exercise 1: Worker Process

Consider the timed automaton for a *Worker* in Figure 1. Assuming that the Worker works for 70 minutes, what is the **minimum and maximum** number of hits (i.e., taking the transition labeled with *hit*) if the worker

(a) starts with a rest (i.e., starts in the state $(\texttt{Rest}, x = 0, y = 0)$), or

(b) is immediately ready to work (i.e., starts in the state $(\texttt{Work}, x = 0, y = 0)$).

Provide valid executions that prove your claims. You may summarize repeated transitions appropriately.

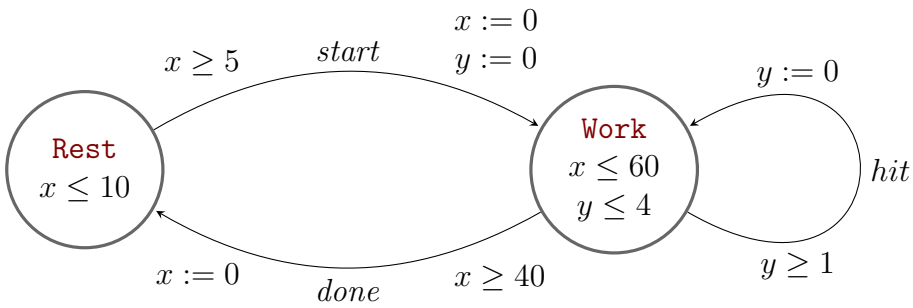

Figure 1: Timed automaton for the worker.

.....Solution sketch .....

a)

Starting in state $(\texttt{Rest}, x = 0, y = 0)$, we have:

min = 11 hits

$(\texttt{Rest}, x = 0, y = 0) \xrightarrow{10} (\texttt{Rest}, x = 10, y = 10) \xrightarrow{start}$
$(\texttt{Work}, x = 0, y = 0) \xrightarrow{4} (\texttt{Work}, x = 4, y = 4) \xrightarrow{hit}$
$(\texttt{Work}, x = 4, y = 0) \xrightarrow{4} (\texttt{Work}, x = 8, y = 4) \xrightarrow{hit}$
$\ldots \xrightarrow{hit} \times 6$
$(\texttt{Work}, x = 32, y = 0) \xrightarrow{4} (\texttt{Work}, x = 36, y = 4) \xrightarrow{hit}$

$(\texttt{Work}, x = 36, y = 0) \xrightarrow{4} (\texttt{Work}, x = 40, y = 4) \xrightarrow{\textit{done}}$
$(\texttt{Rest}, x = 0, y = 4) \xrightarrow{10} (\texttt{Rest}, x = 10, y = 14) \xrightarrow{\textit{start}}$
$(\texttt{Work}, x = 0, y = 0) \xrightarrow{4} (\texttt{Work}, x = 4, y = 4) \xrightarrow{\textit{hit}}$
$(\texttt{Work}, x = 4, y = 0) \xrightarrow{4} (\texttt{Work}, x = 8, y = 4) \xrightarrow{\textit{hit}}$
$(\texttt{Work}, x = 8, y = 0) \xrightarrow{2} (\texttt{Work}, x = 10, y = 2)$

max = 60 hits
$(\texttt{Rest}, x = 0, y = 0) \xrightarrow{5} (\texttt{Rest}, x = 5, y = 5) \xrightarrow{\textit{start}}$
$(\texttt{Work}, x = 0, y = 0) \xrightarrow{1} (\texttt{Work}, x = 1, y = 1) \xrightarrow{\textit{hit}}$
$(\texttt{Work}, x = 1, y = 0) \xrightarrow{1} (\texttt{Work}, x = 2, y = 1) \xrightarrow{\textit{hit}}$
$\dots \xrightarrow{\textit{hit}} \times 57$
$(\texttt{Work}, x = 59, y = 0) \xrightarrow{1} (\texttt{Work}, x = 60, y = 1) \xrightarrow{\textit{hit}}$
$(\texttt{Work}, x = 60, y = 0) \xrightarrow{\textit{done}}$
$(\texttt{Rest}, x = 0, y = 1) \xrightarrow{5} (\texttt{Rest}, x = 5, y = 6)$

b)

Starting in state $(\texttt{Work}, x = 0, y = 0)$, we have:

min = 13 hits
$(\texttt{Work}, x = 0, y = 0) \xrightarrow{4} (\texttt{Work}, x = 4, y = 4) \xrightarrow{\textit{hit}}$
$(\texttt{Work}, x = 4, y = 0) \xrightarrow{4} (\texttt{Work}, x = 8, y = 4) \xrightarrow{\textit{hit}}$
$\dots \xrightarrow{\textit{hit}} \times 6$
$(\texttt{Work}, x = 32, y = 0) \xrightarrow{4} (\texttt{Work}, x = 36, y = 4) \xrightarrow{\textit{hit}}$
$(\texttt{Work}, x = 36, y = 0) \xrightarrow{4} (\texttt{Work}, x = 40, y = 4) \xrightarrow{\textit{done}}$
$(\texttt{Rest}, x = 0, y = 4) \xrightarrow{10} (\texttt{Rest}, x = 10, y = 14) \xrightarrow{\textit{start}}$
$(\texttt{Work}, x = 0, y = 0) \xrightarrow{4} (\texttt{Work}, x = 4, y = 4) \xrightarrow{\textit{hit}}$
$\dots \xrightarrow{\textit{hit}} \times 3$
$(\texttt{Work}, x = 20, y = 0) \xrightarrow{4} (\texttt{Work}, x = 24, y = 4)$

max = 65 hits
$(\texttt{Work}, x = 0, y = 0) \xrightarrow{1} (\texttt{Work}, x = 1, y = 1) \xrightarrow{\textit{hit}}$
$(\texttt{Work}, x = 1, y = 0) \xrightarrow{1} (\texttt{Work}, x = 2, y = 1) \xrightarrow{\textit{hit}}$
$\dots \xrightarrow{\textit{hit}} \times 57$
$(\texttt{Work}, x = 59, y = 0) \xrightarrow{1} (\texttt{Work}, x = 60, y = 1) \xrightarrow{\textit{hit}}$
$(\texttt{Work}, x = 60, y = 0) \xrightarrow{\textit{done}}$
$(\texttt{Rest}, x = 0, y = 1) \xrightarrow{5} (\texttt{Rest}, x = 5, y = 6) \xrightarrow{\textit{start}}$
$(\texttt{Work}, x = 0, y = 0) \xrightarrow{1} (\texttt{Work}, x = 1, y = 1) \xrightarrow{\textit{hit}}$
$\dots \xrightarrow{\textit{hit}} \times 4$
$(\texttt{Work}, x = 5, y = 0)$

Exercise 2: Alarm Timer

An *Alarm Timer* is a timed process which can be set to time-out after a prescribed time period has elapsed. Provide a timed automaton model of an alarm timer T that can be set to time-out after 5, 10, and 30 minutes by receiving a *set* event and reading one of the input values 5, 10, and 30 on the input channel *in*. After the prescribed time period, T signals the time-out by the output action *to*. It is required that the alarm timer can be reset with a new time-out period at any given moment, in particular before the previously set time period has elapsed.

..... Solution sketch .....

There are multiple ways to solve this problem. In this solution, we try to minimize the number of locations. The automaton is given in Figure 2. Observe that the three discrete actions exert the exact same behavior, meaning that we can use a single action to capture all three. We utilize one state variable x (of type clock). The input values (5, 10, or 30) are assigned to *in* and received on channel *set*. The output is transmitted on channel *to*.

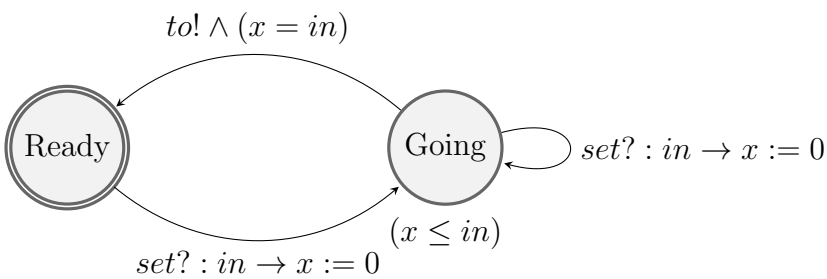

Figure 2: Timed automaton for the alarm timer.

Exercise 3: Fischer's Protocol

Download the UPPAAL model of Fischer's Protocol (FischersProtocol.xml) from Moodle and examine it using the simulator. In addition use verification to check whether the following properties are satisfied or not[1]:

- Does the protocol satisfies the mutual exclusion property:

$$\forall\Box\neg(\texttt{Process(1).Crit} \wedge \texttt{Process(2).Crit})$$

- Does the protocol satisfies the non-starvation property: if a process wants to enter its critical section then it eventually does

$$\texttt{Process(1).Test} \rightarrow \texttt{Process(1).Crit}$$

- Is the protocol deadlock-free, i.e.

$$\forall\Box\neg\text{deadlock}$$

[1] You can find UPPAAL's documentation on queries here: `https://docs.uppaal.org/language-reference/query-syntax/symbolic_queries/`.

Perform verification of the above properties for varying values of the constants D1 and D2. In case the properties does not hold, try to obtain a counter-example. Also, perform the verifications for varying numbers of processes.

.....Solution sketch .....

The first and the third property is satisfied. The second (liveness) property fails to hold for trivial reasons. In fact, it is possible to delay forever in the location-combination (`Process(1).Test`, `Process(2).Test`).

Exercise 4: Worker Process – revisited

Let us revisit the *Worker* process from Exercise 1. Model the Worker in UPPAAL and provide queries to prove your claims of the minimum and maximum hits.

.....Solution sketch .....

First we need to add a counter $hits$ on the hit transition, i.e., we add the update $hits := hits + 1$. Now the number of hits is available to be used in queries. We use the following general query

$$\exists\Diamond\ t = 70 \wedge hits = n$$

The general idea is to provide a value for n such that for a minimum number of hits the query is satisfied for n but not $n - 1$, while for a maximum number of hits the query is satisfied for n but not $n + 1$. The values of n are the onces mentioned in the solution of Exercise 1.

Exercise 5: Symbolic Exploration of Timed Automata

Consider the timed automaton $\mathcal{A}$ in Figure 3 with two clocks x and y. Perform a symbolic zone-base reachability analysis of $\mathcal{A}$. Which of the locations D, E, and F are reachable? Describe the reachable zones upon entry and after delay for the locations A, B, and C.

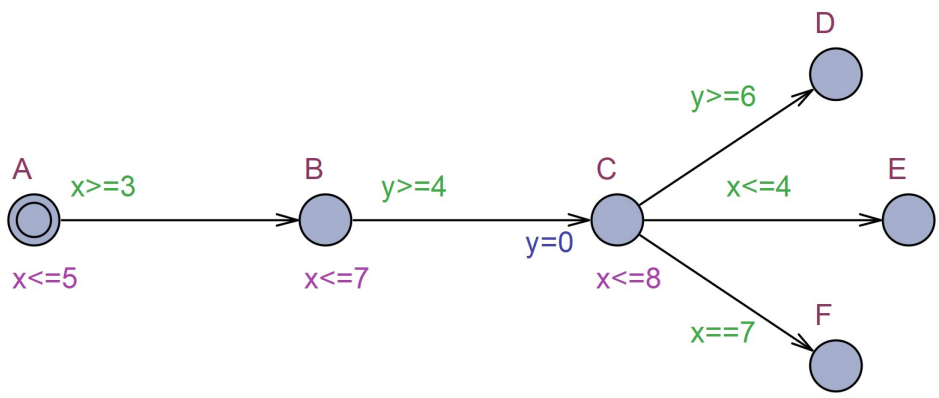

Figure 3: Timed automaton $\mathcal{A}$.

.....Solution sketch .....

Locations E and F are reachable, whereas locations D is not reachable. The symbolic reachable zones of $\mathcal{A}$ are given in Figure 4:

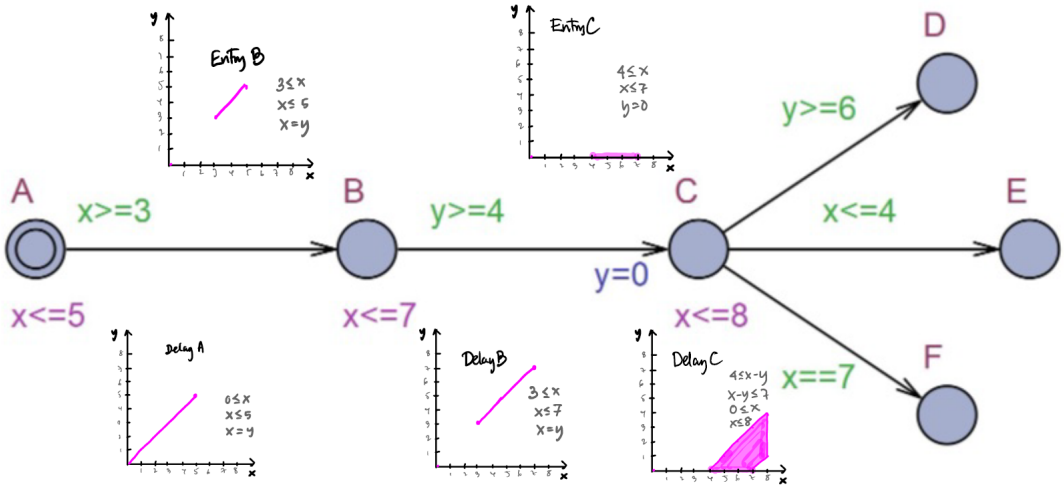

Figure 4: Reachable zones.